AQ3.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

Consider the following statements:

- Statement 1:Statement 1: $0(0,0) + 0(1,1) = (0,0)$0(0, 0) + 0(1, 1) = (0, 0) implies that the set $\{(0,0),(1,1)\}$\{(0, 0), (1, 1)\} is a linearly independent subset of R^2 R2.
- Statement 2:Statement 2: $2(1,0) + 2(0,1) = (2,2)$2(1, 0) + 2(0, 1) = (2, 2) implies that the set $\{(1,0),(0,1),(2,2)\}$\{(1, 0), (0, 1), (2, 2)\} is a linearly dependent subset of R^2 R2.
- Statement 3:Statement 3: $2(1,0) + 2(0,1) = (2,2)$2(1, 0) + 2(0, 1) = (2, 2) implies that the set $\{(1,0),(2,2)\}$\{(1, 0), (2, 2)\} is a linearly dependent subset of R^2 R2.

Find the number of correct statements.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
If $\alpha(1,1,0) + \beta(0,1,1) + \gamma(1,0,1) = \upsilon$α(1, 1, 0) + β(0, 1, 1) + γ(1, 0, 1) = v for some $\alpha, \beta, \gamma \in R$ α, β, γ ∈ R and $\upsilon \in R^3$ v ∈ R3, then choose the set of correct options.

- [ ] If $\upsilon = (0,0,0)$v = (0, 0, 0), then one of the possible solutions of $\alpha, \beta,$ and γα, β, and γ is $\alpha = \frac{1}{2} = \gamma$ α = 21 = γ, and $\beta = -\frac{1}{2}$β = −21.
- [ ] If $\upsilon = (0,0,0)$v = (0, 0, 0), then one of the possible solutions of $\alpha, \beta,$ and γα, β, and γ is $\alpha = \beta = \gamma = 0$ α = β = γ = 0.
- [ ] If $\upsilon = (0,0,0)$v = (0, 0, 0), then the possible solution of $\alpha, \beta,$ and γα, β, and γ is unique.
- [ ] If $\upsilon = (1,0,0)$v = (1, 0, 0), then one of the possible solutions of $\alpha, \beta,$ and γα, β, and γ is $\alpha = \frac{1}{2} = \gamma$ α = 21 = γ, and $\beta = -\frac{1}{2}$β = −21.
- [ ] The set $\{(1,1,0),(0,1,1),(1,0,1)\}$\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\} is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
If $\upsilon = (0,0,0)$v = (0, 0, 0), then one of the possible solutions of $\alpha, \beta,$ and γα, β, and γ is $\alpha = \beta = \gamma = 0$ α = β = γ = 0.
If $\upsilon = (0,0,0)$v = (0, 0, 0), then the possible solution of $\alpha, \beta,$ and γα, β, and γ is unique.
If $\upsilon = (1,0,0)$v = (1, 0, 0), then one of the possible solutions of $\alpha, \beta,$ and γα, β, and γ is $\alpha = \frac{1}{2} = \gamma$ α = 21 = γ, and $\beta = -\frac{1}{2}$β = −21.
The set $\{(1,1,0),(0,1,1),(1,0,1)\}$\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\} is linearly independent.
1 point

Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\}$S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} in R^2 R2. Choose the set of correct options.

- [ ] $\{(1,2),(-1,2)\}$ {(1, 2), (−1, 2)} is a linearly dependent set.
- [ ] $\{(-1,2),(-3,1)\}$ {(−1, 2), (−3, 1)} is a linearly independent set.
- [ ] $\{(-3,1),(-1,2),(3,1)\}$ {(−3, 1), (−1, 2), (3, 1)} is a linearly dependent set.
- [ ] $\{(3,1),(-3,1)\}$ {(3, 1), (−3, 1)} is a linearly dependent set.
- [ ] $\{(3,1),(1,2),(-3,1)\}$ {(3, 1), (1, 2), (−3, 1)} is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-1,2),(-3,1)\}$ {(−1, 2), (−3, 1)} is a linearly independent set.

$\{(-3,1),(-1,2),(3,1)\}$ {(−3, 1), (−1, 2), (3, 1)} is a linearly dependent set.

$\{(3,1),(1,2),(-3,1)\}$ {(3, 1), (1, 2), (−3, 1)} is a linearly dependent set.

1 point

Let VV be a vector space and non zero vectors $v_1, v_2, v_3,$ and $v_4 \in V$v1, v2, v3, and v4 ∈ V . If v_1 v1 is a linear combination of v_i vi, for $i = 2,3,4$i = 2, 3, 4, then which of the following is (are) correct?

- [ ] The set $\{v_1, v_2, v_3, v_4\}$ {v1, v2, v3, v4} cannot be linearly independent.
- [ ] The set $\{v_2, v_3, v_4\}$ {v2, v3, v4} must be linearly dependent.
- [ ] The set $\{v_2, v_3, v_4\}$ {v2, v3, v4} must be linearly independent.
- [ ] The set $\{v_1\}$ {v1} must be linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{v_1, v_2, v_3, v_4\}$ {v1, v2, v3, v4} cannot be linearly independent.

The set $\{v_1\}$ {v1} must be linearly independent.

1 point

Choose the correct set of options.

- [ ] Union of two distinct linearly independent sets may be linearly independent.
- [ ] Union of two distinct linearly independent sets must be linearly dependent.
- [ ] Non-empty intersection of two linearly independent sets must be linearly independent.
- [ ] Non-empty intersection of two linearly independent sets must be linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Union of two distinct linearly independent sets may be linearly independent.

Non-empty intersection of two linearly independent sets must be linearly independent.

If vectors v_1, v_2, v_3 v1, v2, v3 are linearly independent and $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$αv1 + βv2 + γv3 = 0, then find the value of $\alpha + \beta + \gamma$α + β + γ.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point

Consider the set of matrices $S = \{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}, \begin{bmatrix}7\\8\\x\end{bmatrix}\} \subset M_{3\times 1}(R)$S = ⎩ ⎧ ⎡ ⎤ ⎡ ⎤ ⎡ ⎤ ⎫ ⎭ ⎦ ⎣ ⎦ ⎣ ⎦ ⎣ ⎦ ⎣ ⎦ ⎩ ⎭ ⊂ M3×1(R). If $x \in \{9, 4, 2, 0\}$x ∈ {9, 4, 2, 0}, then find the maximum value of xx so that SS is linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

Level 2

1 point
Let SS be a linearly independent subset of R^3 R3. Choose the correct set of options.

- [ ] $S \cup \{(1, 0, 0)\}$S ∪ {(1, 0, 0)} must be linearly independent.
- [ ] $S \cup \{v\}$S ∪ {v} must be linearly independent for any $v \in R^3$ v ∈ R3.
- [ ] $S \setminus \{v\}$S \ {v} must be linearly independent for any $v \in R^3$ v ∈ R3.
- [ ] There may exist some $v \in R^3$ v ∈ R3, such that $S \cup \{v\}$S ∪ {v} is still linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$S \setminus \{v\}$S \ {v} must be linearly independent for any $v \in R^3$ v ∈ R3.
There may exist some $v \in R^3$ v ∈ R3, such that $S \cup \{v\}$S ∪ {v} is still linearly independent.
1 point
Consider a set of vectors $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$S = {(1, 1, −1), (−1, 1, 1), (0, 21, 0), (0, 1, −2), (1, 0, −2)}. Choose the set of correct options.

- [ ] The singleton set $\{(0, 1, -2)\}$ {(0, 1, −2)} is linearly dependent.
- [ ] If $\alpha, \beta \in S$α, β ∈ S and α, βα, β are distinct then $\{\alpha, \beta\}$ {α, β} is a linearly independent set of vectors.
- [ ] The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ {(−1, 1, 1), (0, 21, 0), (0, 1, −2)} is a linearly independent set of vectors.
- [ ] The set SS is a linearly independent set of vectors.
- [ ] The set $\{\alpha, \beta, \gamma\}$ {α, β, γ} is a linearly dependent set of vectors for any $\alpha, \beta, \gamma \in S$α, β, γ ∈ S, where

all the three are distinct vectors.

☐

The set $\{\alpha, \beta, \gamma, \delta\}$ is a linearly independent set of vectors for any $\alpha, \beta, \gamma, \delta \in S$, where all the four are distinct vectors.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

1 point

Let S be the solution set of a system of homogeneous linear equations with 3 variables and 3 equations, whose matrix representation is as follows:

$$Ax = 0$$

where A is the 3×3 coefficient matrix and x denotes the column vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Choose the set of correct options.

☐

If v_1 and v_2 are in S, then any linear combination of v_1 and v_2 will also be in S.

☐

The set S will be a subspace of R^3, with respect to usual addition and scalar multiplication as in R^3.

☐

The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S.

☐

The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in S, then any linear combination of v_1 and v_2 will also be in S.

The set S will be a subspace of R^3, with respect to usual addition and scalar multiplication as in R^3.

The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S.

The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2.

Check Answers

Your score is: 0/10